makes the magnetic field whirl (Figure 9.3). The whirling magnetic field generates an intense electric field like in a dynamo at a hydroelectric power station. The electric field and the whirling magnetic field together fling plasma (hot, ionized gas) upward and downward at near light speed, creating and powering two jets. The jets' directions are held steady (when averaged over years) by the black hole's spin, which is steady due to gyroscopic action.

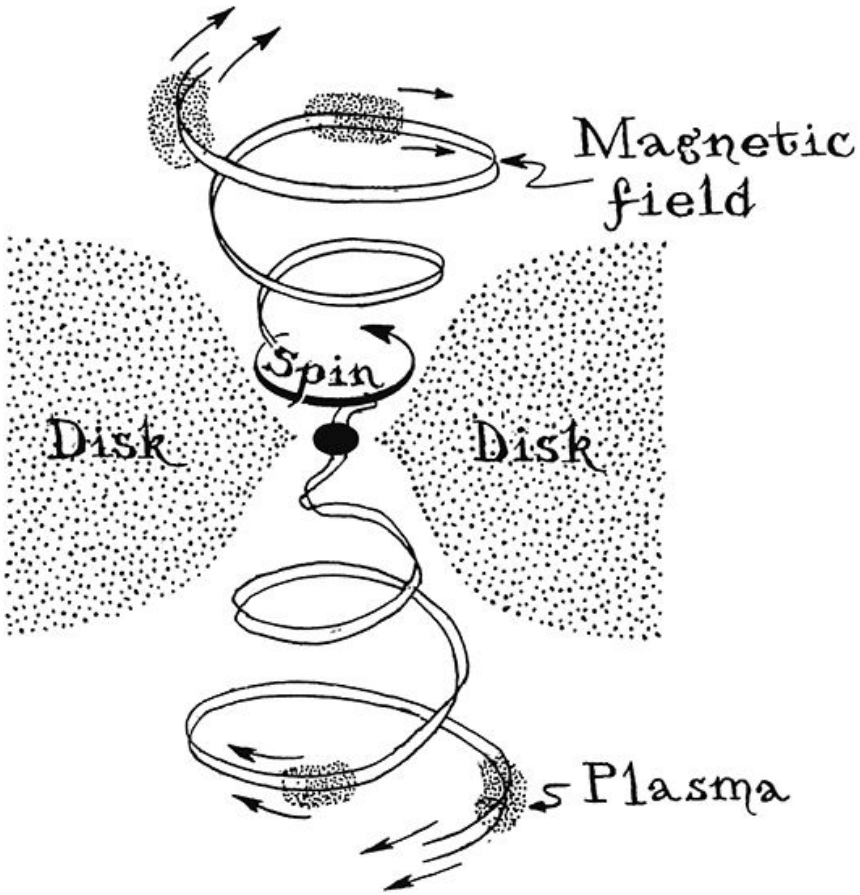


Fig. 9.3. Blandford-Znajek mechanism for generating jets.
[Drawing by Matt Zimet based on a sketch by me; from my book Black Holes & Time Warps: Einstein's Outrageous Legacy.*]*

In 3C273 only one jet was bright enough to see (Figure 9.1), but in many other quasars both are seen.